

Joshua Aney

Rochester, MN | josh.aney@icloud.com | github.com/joshaney324 | joshaney324.github.io

EDUCATION

Montana State University

Bachelor of Science, Computer Science, GPA: 3.96

Bozeman, MT

May 2026

HONORS & AWARDS

President's List: Spring 2022; Spring 2023, Fall 2023; Spring 2024; Spring 2025, Fall 2025; Spring 2026, Fall 2026

Dean's List: Fall 2022; Fall 2024

RESEARCH EXPERIENCE

Research Technologist

June 2026 – Present

CaMP Lab (Cybernetics and Motor Physiology), Mayo Clinic

Rochester, MN

- **Tools:** MATLAB, Signal Processing, Adobe Illustrator, Linear Algebra, Technical Writing
- Previously Summer Research Intern in the same lab, 2024 and 2025.
- Conducted algorithmic analysis on human sEEG data to study how the brain responds to auditory stimuli.
- Implemented signal processing pipelines including noise reduction, baseline correction, and artifact rejection.
- Applied and modified Canonical Response Parameterization to allow for better temporal feature extraction when applied to auditory evoked responses.
- Created a new method for dynamic brain network identification using evoked responses.
- Leading development of a reference and preprocessing guide for general sEEG data analysis.

Undergraduate Research Assistant

January 2026 – June 2026

HACR Lab (Harnessing Automation in Cybersecurity Reasoning), Montana State University

Bozeman, MT

- **Tools:** PyTorch, Ghidra, High Performance Computing, SLURM
- Built pipelines to produce control flow graphs from compiled binary packages for code embedding generation.
- Designed Graph Neural Network architectures using PyTorch based on recent research publications.
- Ran machine learning workloads on Tempest, Montana State University's high performance computing system.

TEACHING & CURRICULUM DEVELOPMENT

Independent Study – CSCI 491-7 Hypermedia Systems

Fall 2025

Montana State University

Bozeman, MT

- **Tools:** Java, SQLite, HTMX, HTML/CSS, MVC Architecture, Database Design
- Collaborated with a professor and a team of three students to design and develop a semester-long project template for an advanced web development course teaching a hypermedia-first approach using server-side rendering with HTMX.
- Implemented core website functionality, including the MVC controller layer, authentication and authorization systems, and multi-step business workflows, for a veterinary clinic management system.
- Designed a database schema with 11 interconnected tables, including polymorphic relationships and conversation-based messaging, to teach role-based access control, pagination, and transactional integrity.
- Structured a progressive assignment sequence so students could implement new controllers by following established patterns in the codebase.

PUBLICATIONS & PRESENTATIONS

Papers in Preparation

- Aney, J., et al. *Response Lag Correlation: Identifying response networks through evoked response lags.*
- Aney, J., et al. *A Reference and Preprocessing Guide for sEEG Data.*

Conference Abstracts

- Aney, J., Jensen, M., Kerezoudis, P., Canales-Johnson, A., Hermes, D., Baker, M., Miller, K. *Parameterization of intracranial auditory evoked responses reveals cortical novelty and attention effects*. BRAIN Initiative Conference, 2026.
- Aney, J., Jensen, M., Kerezoudis, P., Canales-Johnson, A., Hermes, D., Baker, M., Miller, K. *Parameterization of intracranial auditory evoked responses reveals cortical novelty and attention effects*. GATTLE, Mayo Clinic, 2026.

Posters

- Aney, J., Jensen, M., Kerezoudis, P., Canales-Johnson, A., Hermes, D., Baker, M., Miller, K. *Parameterization of intracranial auditory evoked responses reveals cortical novelty and attention effects*. Poster presented at BRAIN Initiative Conference, 2026.

SKILLS & COURSEWORK

Languages: Python, MATLAB, Java, C, SQL, C#

Frameworks: PyTorch, TensorFlow, WebDataset, .NET Core, SLURM, NumPy, Matplotlib

Tools: Linux, JetBrains Products, GitHub, VSCode, Adobe Illustrator, Overleaf, Claude Code

Coursework: Machine Learning, Linear Algebra, Artificial Intelligence, Greedy Algorithms, Flow Networks, Optimization, Linear and Integer Linear Programming, Technical Writing

TECHNICAL PROJECTS

- | | |
|---|-----------|
| Machine Learning and AI Models from Scratch <i>Python, NumPy, Linear Algebra, Latex</i> | 2024-2026 |
| <ul style="list-style-type: none"> • Implemented the following machine learning algorithms from scratch: Naïve Bayes, K-Nearest Neighbors, K-Means, and DBSCAN, Multi-Layer-Perceptron, Q-Learning, SARSA, and Value Iteration • Implemented a First-Order-Logic engine to perform inference on knowledge bases using resolution. • Implemented the following optimization algorithms from scratch: Backpropagation, Genetic Algorithm, Differential Evolution, Particle Swarm Optimization, and Simulated Annealing | |
| Bozeman Bike Trails Application <i>C#, Razor Pages, SQL Server, Mapbox</i> | 2024 |
| <ul style="list-style-type: none"> • Developed a web application to track user-ridden bike trail GPS data | |
| Hand Tracking / ASL Alphabet Detection <i>Python, TensorFlow, MediaPipe</i> | 2024 |
| <ul style="list-style-type: none"> • Created a pipeline to collect data, train a dense neural network, and provide real time predictions for video inputs. | |

OTHER ACTIVITIES

- | | |
|--|--|
| President
<i>Montana State University Tennis Club</i> | August 2023 – Present
Bozeman, MT |
| Section Leader
<i>Montana State University Drumline</i> | April 2025 – January 2026
Bozeman, MT |
| Tennis Coach
<i>Rochester Tennis Connection</i> | Summers 2023, 2025
Rochester, MN |
| Substitute Tennis Coach
<i>Bobcat Anderson Tennis Center, Montana State University</i> | 2022 – 2026
Bozeman, MT |